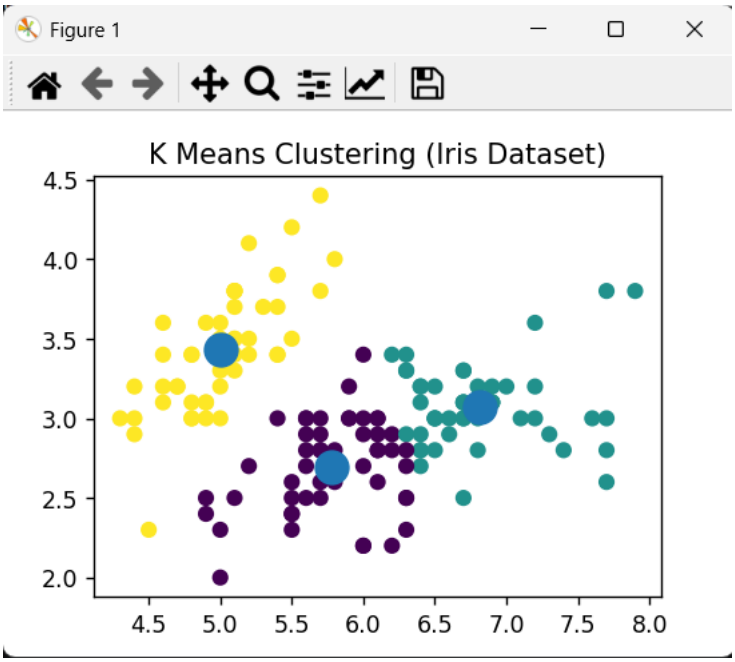
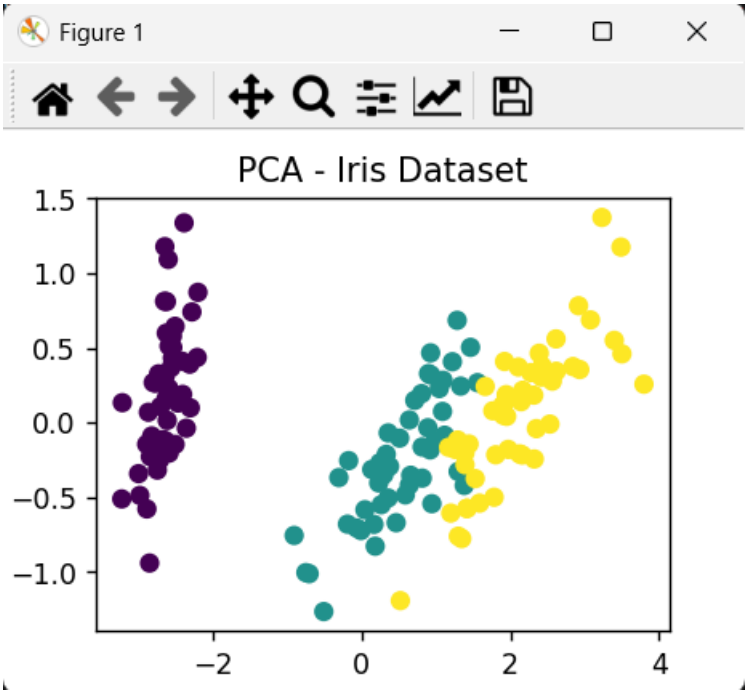
```
# K MEANS
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

data=load_iris()
x=data.data[:,:2]

kmeans=KMeans(n_clusters=3,random_state=0)
y_kmeans=kmeans.fit_predict(x)

plt.scatter(x[:,0],x[:,1],c=y_kmeans)
plt.scatter(kmeans.cluster_centers_[:,0],kmeans.cluster_centers_[:,1],s=200)
plt.title("K Means Clustering (Iris Dataset)")
plt.show()

# PCA
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

data = load_iris ()
X = data.data
y = data.target

pca = PCA(n_components=2)
X_pca = pca.fit_transform (X)

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.title ("PCA - Iris Dataset")
plt.xlabel ("PC1")
plt.ylabel ("PC2")
plt.show ()
```

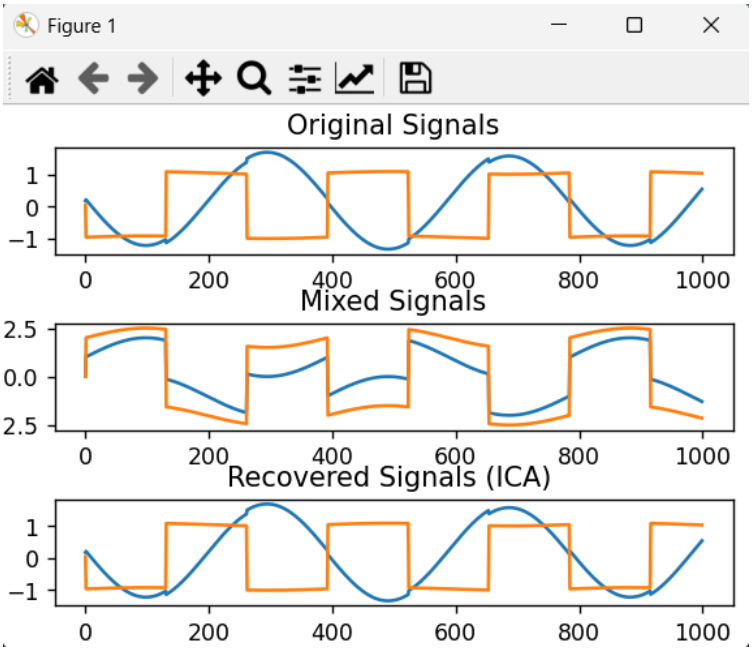
```
#ICA
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import FastICA

time = np.linspace (0, 8, 1000)

s1 = np.sin (2 * time)
s2 = np.sign(np.sin(3 * time))

S = np.c_[s1, s2]

A = np. array([ [1, 1], [0.5, 2]] )
X = np.dot (S, A.T)

ica = FastICA(n_components=2)
S = ica.fit_transform (X)

plt.figure ()
plt.subplot (3,1,1)
plt.title ("Original Signals")
plt.plot (S)

plt.subplot (3,1,2)
plt.title ("Mixed Signals")
plt.plot (X)

plt.subplot (3,1,3)
plt.title("Recovered Signals (ICA) ")
plt.plot (S )

plt.tight_layout ()
plt.show ()
```

```
# Apriori
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

dataset = [
['Milk', 'Bread', 'Butter'],
['Bread', 'Butter' ],
['Milk', 'Bread'],
['Milk', 'Butter'],
['Bread', 'Butter' ]
]

from mlxtend.preprocessing import TransactionEncoder
te = TransactionEncoder ()
te_data = te.fit (dataset) .transform (dataset)
df = pd. DataFrame (te_data, columns=te.columns_)

frequent_items = apriori (df, min_support=0.4, use_colnames=True)
rules = association_rules (frequent_items, metric="confidence", min_threshold=0.5)

print ("Frequent Itemsets:\n", frequent_items)
print ("\nAssociation Rules: \n", rules)
```

Outputs:








```
(.venv) PS D:\Internship\Day51> python task.py
Frequent Itemsets:
   support                 itemsets
0      0.8          frozenset({Bread})
1      0.8         frozenset({Butter})
2      0.6           frozenset({Milk})
3      0.6  frozenset({Bread, Butter})
4      0.4    frozenset({Bread, Milk})
5      0.4   frozenset({Butter, Milk})

Association Rules:
            antecedents          consequents  antecedent support  consequent support  support  confidence  ...  leverage  conviction  zhangs_metric  jaccard  certainty  kulczynsk
i
0   frozenset({Bread})  frozenset({Butter})                 0.8                 0.8      0.6    0.750000  ...     -0.04         0.8      -0.250000      0.6  -0.250000    0.750000
1  frozenset({Butter})   frozenset({Bread})                 0.8                 0.8      0.6    0.750000  ...     -0.04         0.8      -0.250000      0.6  -0.250000    0.750000
2   frozenset({Bread})    frozenset({Milk})                 0.8                 0.6      0.4    0.500000  ...     -0.08         0.8      -0.500000      0.4  -0.250000    0.583333
3    frozenset({Milk})   frozenset({Bread})                 0.6                 0.8      0.4    0.666667  ...     -0.08         0.6      -0.333333      0.4  -0.666667    0.583333
4  frozenset({Butter})    frozenset({Milk})                 0.8                 0.6      0.4    0.500000  ...     -0.08         0.8      -0.500000      0.4  -0.250000    0.583333
5    frozenset({Milk})  frozenset({Butter})                 0.6                 0.8      0.4    0.666667  ...     -0.08         0.6      -0.333333      0.4  -0.666667    0.583333

[6 rows x 14 columns]
```